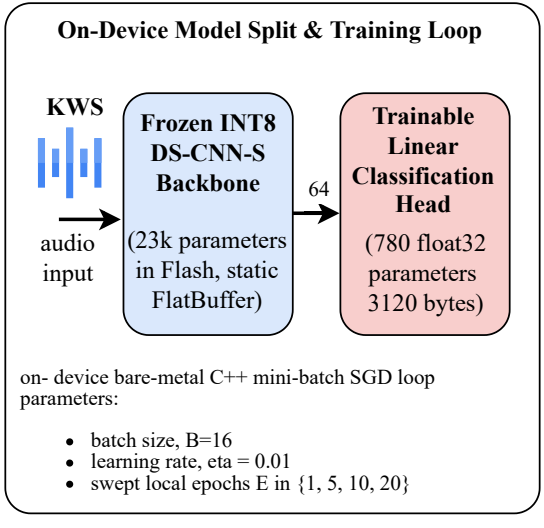
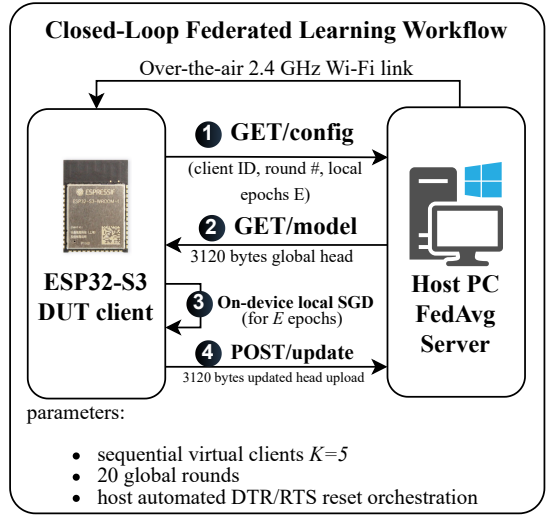

On-Device Model Split & Training Loop
KWS
audio input
Frozen INT8 DS-CNN-S Backbone
(23k parameters in Flash, static FlatBuffer)
64
Trainable Linear Classification Head
(780 float32 parameters 3120 bytes)
on- device bare-metal C++ mini-batch SGD loop parameters:
batch size, B=16
learning rate, eta = 0.01
swept local epochs E in {1, 5, 10, 20}


Closed-Loop Federated Learning Workflow
Over-the-air 2.4 GHz Wi-Fi link
1 GET/config
(client ID, round #, local epochs E)
2 GET/model
3120 bytes global head
3 On-device local SGD
(for E epochs)
4 POST/update
3120 bytes updated head upload
ESP32-S3 DUT client
Host PC FedAvg Server
parameters:
sequential virtual clients K=5
20 global rounds
host automated DTR/RTS reset orchestration